

JSS MAHAVIDHYAPEETHA

JSS SCIENCE AND TECHNOLOGY UNIVERSITY

SRI JAYACHAMARAJENDRA COLLEGE OF ENGINEERING



- Constituent College of JSS Science and Technology University
- Approved by A.I.C.T.E
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DEPARTMENT OF PHYSICS

TUTORIAL PROBLEMS

UNIT-1: QUANTUM MECHANICS & COMPUTING

Exercise Problems:

1. A microscope using photons is employed to locate an electron in an atom within a distance of $0.2 \text{ \AA}$. What is the uncertainty in the momentum of the electron located in this way? What is the uncertainty in velocity?
2. Calculate the uncertainty in the velocity of α -particle which is located within $4 \text{ \AA}$.
3. A proton is confined to a nucleus of radius 10^{-14} m . Calculate the minimum uncertainty in its momentum and also calculate minimum kinetic energy of proton.
4. A typical electron sailing down the beam of a TV picture tube has energy of about $2 \times 10^{-15} \text{ J}$, and uncertainty in its position is 10^{-10} m . Calculate the corresponding uncertainty in its speed and determine the fractional uncertainty in speed.
5. An electron of energy 250 eV is passed through a circular hole of radius 10^{-6} m . What is the uncertainty introduced in the angle of emergence?
6. The position and momentum of an electron with energy of 1 keV are simultaneously determined. If the inherent uncertainty in the measurements of its position is $1 \text{ \AA}$, what is the minimum percentage uncertainty in its momentum?
7. The inherent uncertainty in the measurement of time spent by iridium-191 nuclei in the excited state is found to be $1.4 \times 10^{-10} \text{ s}$. Estimate the uncertainty that results in its energy in the excited state.
8. What is the uncertainty in the frequency of a photon whose life time is about 10^{-8} s ?
9. A spectral line of wavelength $5461 \text{ \AA}$ has a width of $10^{-4} \text{ \AA}$. Evaluate the minimum time spent by the electrons in the upper energy state between the excitation and de-excitation processes.

10. What is the uncertainty in the location of a photon if its wave length of $5000 \text{ \AA}$ is known to an accuracy of one part in 10^7 ?
11. Calculate the uncertainty in the position of electron which has been accelerated through a potential difference of $1000 \pm 6 \text{ V}$.
12. On the basis of the uncertainty principle, estimate the minimum energy of an electron that is confined in a cubic cavity of volume $50 \times 50 \times 50 \text{ \AA}^3$.
13. A particle like electron is bound in a one dimensional potential well of width $1 \text{ \AA}$, but of infinite height. Find its energy values in the ground state, and also in the first two excited states.
14. A particle is confined to one dimensional potential well of width 0.2 nm . It is found that when the energy of the particle is 230 eV , its Eigen function has 5 antinodes. Find the mass of the particle.
15. An electron is confined to move between two rigid walls separated by 10^{-9} m . Calculate the de- Broglie wavelength representing the first three allowed energy states of electron and also calculate the corresponding energies.
16. An electron trapped in a one dimensional potential well of width $1 \text{ \AA}$. What is the probability of finding the electron in the region from $x = 0.09 \times 10^{-10} \text{ m}$ to $x = 0.11 \times 10^{-10} \text{ m}$ in the ground state.
17. A quantum particle confined to one dimensional potential box of width 'a' is in its first excited state. What is the probability of finding the particle over an interval of 'a/2' marked symmetrically at the center of the box.
18. A particle is moving in one dimensional potential box of width $10 \text{ \AA}$. Calculate the probability of finding the particle with an interval of $1 \text{ \AA}$ at the center of the box when it is in its state of least energy.

Assignment Problems:

19. The natural uncertainty in the measurement of speed of an electron in an atom is estimated to be 2.2×10^4 m/s conceding the ideal set up and error free measurement. Estimate the minimum width about which the electron stays confined in the atom.
20. Calculate the energy of an electron in a hydrogen atom using Heisenberg's uncertainty principle (Given: Bohr radius = $0.529 \text{ \AA}$).
21. In a measurement that involved an inherent uncertainty of 0.003%, the speed of an electron was found to be 800 m/s. Calculate the corresponding uncertainty involved in the position determination.
22. An electron is trapped in a one dimensional potential well of width $1 \text{ \AA}$. Find the amount of energy required to excite the electron to its first and second excited states.
23. A particle limited to the x-axis has the wave function $\psi = ax$ between $x = 0$ and $x = 1$; $\psi = 0$ elsewhere. Find the probability that the particle can be found between $x = 0.45$ and $x = 0.55$.
24. Some organic dye molecules have a chain of carbon atoms along a line. An electron participating in the bonding of these atoms behaves somewhat as a particle in a box. In a model of one type of dye molecule, suppose that an electron is confined to a chain of length 0.94 nm , then:
 - (a) Determine the energy of the four lowest states.
 - (b) Compare the probability densities of at the midpoint of the region for the $n = 3$ and $n = 4$ states.
 - (c) The color of the dye is due to the transition between these states. What is the color?
25. A particle is moving in one dimensional potential well of width $25 \text{ \AA}$. Assuming the particle is in its least state of energy, calculate the probability of finding the particle in an interval of $5 \text{ \AA}$ at distances of $a/2$, $a/3$ and at a , where 'a' is the width of the well.